Initial Value Ordinary Differential Equations

Chapter 6 of Numerical Analysis

Yiping Cheng
ypcheng@bjtu.edu.cn
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School of Electronic and Information Engineering
Beijing Jiaotong University

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Problem Formulation

Ordinary differential equation of first order

In this chapter, we consider the differential equation

$$y'(x) = f(x, y(x))$$

where $f(x, y)$ is a given function.

Remark

- This equation is called an **ordinary differential equation** (abbreviated as ODE) because there is only one independent variable x; otherwise partial derivatives will be involved and the equation will be called a partial differential equation.
- This equation is of **first order** because no second or higher-order derivatives of y occur in the equation.
- For pedagogical purposes, in this chapter the discussion is restricted to the case that y **is a scalar** function. However, most of the results can be generalized to the vector case in a natural manner.

Initial value problem for ODE

It was observed very early by Newton, Leibniz and Euler that the solution usually contains a free parameter, so that it is uniquely determined only when an initial value $y(x_0) = y_0$ is prescribed. The pair of equations

$$\begin{aligned} y'(x) &= f(x, y(x)) \\ y(x_0) &= y_0 \end{aligned}$$

is known as an "**initial value problem of an ordinary differential equation**" (hereafter abbreviated as IVODE). Our interest in this chapter is with exactly this problem, where the aim is to obtain approximate values of $y(x)$ for specific values of x, usually with $x > x_0$, corresponding to the prediction of the future states of a differential equation system.

The case of an integration problem

When $f(x, y)$ is only a function of x only, then the IVODE degenerates into an integration problem:

$$\begin{aligned} y'(x) &= f(x) \\ y(x_0) &= y_0. \end{aligned}$$

Its solution is simply

$$y(x) = y_0 + \int_{x_0}^{x} f(t)dt.$$

And if an analytic solution is not available, then the numerical methods described in the last chapter can be employed.

The case of an autonomous system

When $f(x, y)$ is only a function of y only, then the IVODE reduces into an autonomous system:

$$\begin{aligned} y'(x) &= f(y(x)) \\ y(x_0) &= y_0. \end{aligned}$$

For autonomous systems, some methods (analytical or numerical) can be simplified a bit as compared to non-autonomous systems.

Logistic population growth – an example IVODE

Imagine we have bacteria growing in a petri dish. Let t denote time. The number of bacteria in the dish $y(t)$ is governed by the following equations:

$$\begin{aligned} y'(t) &= 0.294(1 - \frac{y(t)}{10000})y(t) \\ y(0) &= 1000. \end{aligned}$$

It can be shown that the solution is

$$y(t) = \frac{10^7}{9000e^{-0.294t} + 1000}.$$

This is an example where an analytical solution can be obtained. But for the majority of IVODEs, an analytical solution is not feasible, and we have to rely on numerical methods.

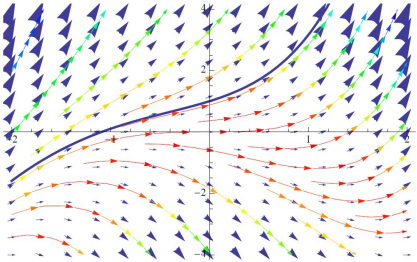
Direction fields

A direction field is to graphically represent solutions to a first-order differential equation. At each point in a direction field, a line segment appears whose slope is equal to the slope of a solution to the differential equation passing through that point. Let us now draw a direction field of the ODE

$$y'(t) = t^2 + y(t), \quad 0 < t < 4.$$

At each point (t, y) we draw the line with slope $t^2 + y$; this is illustrated in the figure below where numerous streamlines have been sketched. To thread a solution through the direction field start at a point and follow the solution, remembering that solutions don't cross and that nearby tangent lines should be nearly the same.

To see which streamline corresponds to the solution with $y(0) = 1$ we locate the point $(0, 1)$ and follow the tangents; this solution is indicated by a thick black line in the direction field plot below. If a different initial condition is imposed, then we get a different streamline.



Numerical solution of IVODEs

When an analytical solution is not feasible, we must turn to a numerical approximation to the solution where we give up finding a formula for the solution at all times and instead find an approximation at a set of distinct times, called **mesh points**. Once the approximate solution is obtained at the points, the approximate solution at other points in the interval can be found by interpolation.

In this chapter, unless otherwise stated, we stipulate that **the mesh points are equally distributed** throughout the interval $[a, b]$ with $a = x_0$. This condition is ensured by choosing a positive integer N and selecting the mesh points $x_i = a + ih$, for each $i = 0, 1, \ldots, N$. The common distance between the points $h = (b - a)/N = x_{i+1} - x_i$ is called the **step size**.

However, many professional solvers are able to use variable step size, where the step size is adjusted according to the local behaviour of the function f.

The strategy to approximate the solution to an IVODE at each point x_i is to use the given initial information at x_0, i.e., y_0 and the slope evaluated at x_0 (i.e., $f(x_0, y_0)$), to get an approximate at x_1. Then the information at x_1 and possibly x_0 is used to get an approximation at x_2. This process is continued until we have an approximation at the final time $x = b$. Each method that we consider has a different way to approximate the solution at the next time step.

A survey of numerical methods for IVODEs

The standard methods for approximating the solution of an IVODE fall into two broad categories which are **one-step** (or **single-step**) methods and **linear multistep** methods. Each category consists of families of **explicit** methods and families of **implicit** methods of varying degrees of accuracy. A single-step method only uses information from one previously computed solution (i.e., at x_i) to compute the solution at x_{i+1}; whereas a linear multistep method uses information at several previously calculated points; for example, a two-step method uses information at x_i and x_{i-1} to predict the solution at x_{i+1}. Using previously calculated information is how linear multistep methods improve on the linear accuracy of the Euler methods. One-step methods improve the accuracy by performing intermediate approximations in $(x_i; x_{i+1}]$ which are then discarded. There are advantages and disadvantages to each class of methods.

Some notation

The following are tacitly assumed throughout this chapter:

$$y'(x) = f(x, y(x)) \text{ for all } x \tag{1}$$

$$y_0 = y(x_0) \tag{2}$$

$$x_0 = a, h > 0, N = \lfloor \frac{b-a}{h} \rfloor, x_i = x_0 + ih, \text{ for } i = 1, \dots, N. \tag{3}$$

We need different notation for the approximated values at each time step x_i from the exact solution to the IVODE. In the sequel, we use **small case letters** for the exact solution to the IVODE (such as $y(x)$, $p(x)$, etc.) and **capital letters** for the approximation. To be specific, we denote by Y_i the approximated value of $y(x_i)$ for $i = 1, \dots, N$. We know the given value of y at x_0, so we set $Y_0 = y_0$.

One-Step Methods

Mathematical description of one-step methods

A general mathematical description of one-step methods is

$$Y_0 = y_0 \tag{4}$$

$$Y_{i+1} = \Gamma(x_i, Y_i; h),\ i = 0, 1, \dots, N-1. \tag{5}$$

Let

$$\Phi(x, y; h) = \frac{\Gamma(x, y; h) - y}{h}. \tag{6}$$

Then (5) is equivalent to

$$Y_{i+1} = Y_i + h\Phi(x_i, Y_i; h),\ i = 0, 1, \dots, N-1. \tag{7}$$

This will be the basis for our later derivation.

Explicit methods and implicit methods

If Φ in (7) (or equivalently, Γ in (5)) is an explicit function, then the method is called an explicit one. A non-explicit method is called an implicit method. In an implicit method, there are quantities to be computed that appear on both sides of a nonlinear equation thus we need iterations to find the quantities.

As will be clear later, an advantage of implicit methods is better stability and thus more suitable for stiff ODEs. A disadvantage of implicit methods is higher computation cost per step.

Convergence of a method

Definition

A numeric method (7) is said to converge to IVODE (1), if with assumption (3),

$$\lim_{h\to 0, Y_0\to y(x_0)} \max_{1\le i\le N} |y(x_i) - Y_i| = 0. \tag{8}$$

We do not suppose that the scheme has the exact initial condition of the ODE. Indeed, on one hand, there could be a truncation error in the initial condition. On the other hand, the initial condition may not be known exactly.

Generally, just as the solution of a differential system is continuous with respect to the initial data, the solution approximated by a one-step scheme must be continuous with respect to a perturbation of the initial conditions.

Convergence, as we will see, results from two properties. The first, stability, is a property of the scheme. This ensures that the scheme does not amplify too much the errors created at each step. The other, consistency, describes a relation between the scheme and the differential system. It implies that the scheme does not differ much from the solution locally.

Zero-stability of a method

Definition

A numeric method (7) is said to be zero-stable if with assumption (3), there exist h^* and a constant M such that, for all Y_0, for all Z_0, for all $h \leq h^*$, and for every sequence of vectors ϵ_i, the sequences Y_i and Z_i defined by the relations

$$\begin{aligned} Y_{i+1} &= Y_i + h\Phi(x_i, Y_i; h) \\ Z_{i+1} &= Z_i + h\Phi(x_i, Z_i; h) + \epsilon_i \end{aligned} \tag{9}$$

satisfy the inequality

$$|Y_i - Z_i| \leq M(|Y_0 - Z_0| + \sum_{k=0}^{i-1} |\epsilon_k|), \quad \forall i \leq N(h). \tag{10}$$

Note: The notation $N(h)$ is to stress that N depends on h.

Local truncation error and consistency of a method

Definition (local truncation error)

The local truncation error of numeric method (7) for solving (1) is defined as

$$\tau(x;h) = y(x+h) - y(x) - h\Phi(x, y(x); h). \tag{11}$$

Definition (consistency)

Numeric method (7) is said to be consistent with (1), if with assumption (3),

$$\lim_{h\to 0} \sum_{i=0}^{N(h)-1} \underbrace{|\, y(x_i+h) - y(x_i) - h\Phi(x_i, y(x_i); h)\,|}_{\tau(x_i;h)} = 0. \tag{12}$$

Convergence = zero stability + consistency

Theorem

A numeric method (7) converges to IVODE (1), if it is zero-stable and consistent with (1).

Proof.

We let $Z_i = y(x_i)$. Then

$$Z_{i+1} - Z_i - h\Phi(x_i, Z_i; h) = \epsilon_i$$

is the local truncation error $\tau(x_i; h)$. We can apply the inequality (10) and we have

$$|Y_i - y(x_i)| \leq M(|Y_0 - y(x_0)| + \sum_{k=0}^{i-1} |\epsilon_k|).$$

From the consistency hypothesis, we see that $Y_i - y(x_i)$ tends uniformly to zero with respect to i as h tends to 0. □

A sufficient condition for zero stability

Theorem

For a method to be zero-stable, it is sufficient that there exists h^* and a constant Λ such that

$$|\Phi(x, y; h) - \Phi(x, z; h)| \leq \Lambda |y - z|, \quad \forall x \in [a, b], \forall y, \forall z, \forall h \in (0, h^*]. \tag{13}$$

Proof.

From (9) if we subtract Z_{i+1} from Y_{i+1} and apply the triangle inequality, we get

$$|Y_{i+1} - Z_{i+1}| \leq |Y_i - Z_i| + h|\Phi(x_i, Y_i; h) - \Phi(x_i, Z_i; h)| + |\epsilon_i|.$$

From hypothesis (13), we therefore have the inequality

$$|Y_{i+1} - Z_{i+1}| \leq (1 + h\Lambda)|Y_i - Z_i| + |\epsilon_i|.$$

We can therefore obtain

$$\begin{aligned} |Y_i - Z_i| &\leq (1 + h\Lambda)^i |Y_0 - Z_0| + \sum_{k=0}^{i-1} (1 + h\Lambda)^{i-1-k} |\epsilon_k| \\ &\leq (1 + h\Lambda)^i (|Y_0 - Z_0| + \sum_{k=0}^{i-1} |\epsilon_k|) \\ &\leq e^{(b-a)\Lambda} (|Y_0 - Z_0| + \sum_{k=0}^{i-1} |\epsilon_k|). \end{aligned}$$

□

Necessary and sufficient condition for consistency

Theorem

Let f be a continuous function of $x \in [a, b]$, y, and $h \in [0, h^*]$. A necessary and sufficient condition so that the scheme (7) is consistent with system (1) is that

$$\Phi(x, y; 0) = f(x, y) \tag{14}$$

A proof is omitted because it is too complicated. For a proof see page 389, Numerical Analysis – A Mathematical Introduction, by Michelle Schatzman.

The forward Euler method

The forward Euler method is the simplest one-step method. It is probably the best known method for numerical solution to the IVODE. This method was named after Leonhard Euler and is often just referred to as "the Euler method". It can be derived that

$$y(x_{i+1}) = y(x_i) + \int_{x_i}^{x_{i+1}} f(t, y(t))dt.$$

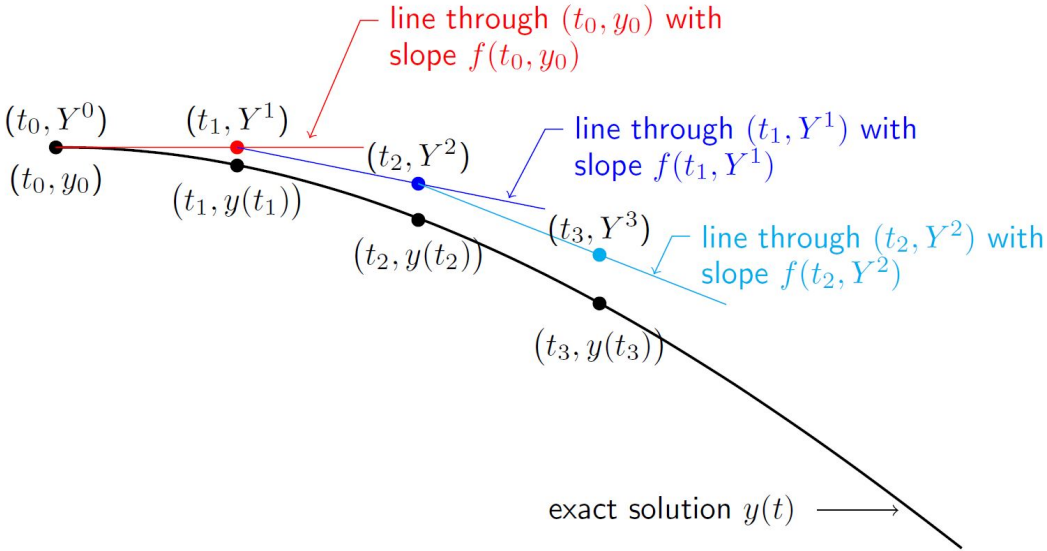
Applying the left-hand rectangle rule to the interval $[x_i, x_{i+1}]$, we obtain

$$y(x_{i+1}) \approx y(x_i) + hf(x_i, y(x_i)),$$

which leads to the following **forward Euler** formula:

$$Y_{i+1} = Y_i + hf(x_i, Y_i), \ i = 0, 1, \ldots, N-1. \tag{15}$$

Obviously, this is an **explicit method**.


line through (t_0, y_0) with slope $f(t_0, y_0)$
(t_0, Y^0)
(t_1, Y^1)
(t_0, y_0)
$(t_1, y(t_1))$
(t_2, Y^2)
line through (t_1, Y^1) with slope $f(t_1, Y^1)$
(t_3, Y^3)
$(t_2, y(t_2))$
line through (t_2, Y^2) with slope $f(t_2, Y^2)$
$(t_3, y(t_3))$
exact solution $y(t)$

The backward Euler method

Again we start from

$$y(x_{i+1}) = y(x_i) + \int_{x_i}^{x_{i+1}} f(t, y(t))dt.$$

Applying the right-hand rectangle rule to the interval $[x_i, x_{i+1}]$, we obtain

$$y(x_{i+1}) \approx y(x_i) + hf(x_{i+1}, y(x_{i+1})),$$

which leads to the following **backward Euler** formula:

$$Y_{i+1} = Y_i + hf(x_{i+1}, Y_{i+1}),\ i = 0, 1, \ldots, N-1.$$

Obviously, this is an **implicit method**.

Local truncation error

For an IVODE (1–3) and a one-step method (4–7), the local truncation error at the i-th step is defined as

$$\tau_i = y(x_{i+1}) - y(x_i) - h\Phi(x_i, y(x_i); h). \tag{16}$$

Hence, local truncation error is the error

$$y(x_{i+1}) - Y_{i+1}$$

assuming that $Y_i = y(x_i)$. It is a mathematical concept and does not exist in practice.

Local truncation error of the forward Euler method

The local truncation error at the i-th step of the forward Euler method is

$$\tau_i = y(x_{i+1}) - y(x_i) - hf(x_i, y(x_i)). \tag{17}$$

Suppose that $|y''(x)| \leq M$ for all x. Then for all $i = 0, \ldots, N-1$, by Taylor's theorem, there is $\xi_i \in (x_i, x_{i+1})$ such that

$$y(x_{i+1}) = y(x_i) + hf(x_i, y(x_i)) + \frac{h^2}{2} y''(\xi_i).$$

And

$$\tau_i = \frac{h^2}{2} y''(\xi_i),$$

$$|\tau_i| \leq \frac{M}{2} h^2.$$

This says that the local truncation error of the forward Euler method is of order 2.

Remark

The existence of an upper bound of $|y''|$ can be pre-guaranteed by the behavior of f. For example, suppose that $f(x, y)$ is differentiable everywhere and $f(x, y)$, $f_x(x, y)$, and $f_y(x, y)$ are all bounded on the xy-plane. Since

$$y''(x) = \frac{df(x, y(x))}{dx} = f_x(x, y(x)) + f_y(x, y(x))y'(x) = [f_x + f_y f](x, y(x)),$$

we see that y'' is also bounded.

Local truncation error of the backward Euler method

The local truncation error of the backward Euler method is also of order 2. Its proof is rather tricky so is omitted here. It can be found at

https://math.stackexchange.com/questions/1750267/local-truncation-error-of-implicit-euler

Global error

For an IVODE (1–3) and a one-step method (4–7), the global error is defined as

$$e_i = |y(x_i) - Y_i| \tag{18}$$

$$E = \max_{1 \le i \le N} e_i \tag{19}$$

This is a measure of how well the approximation Y_i agrees with the true solution over the whole interval.

Relationship between local truncation error and global error

Theorem

Suppose that there are $T > 0$ and $p > 0$ such that for $i = 0, 1, \dots, N-1$, $|\tau_i| \le T h^{p+1}$. Suppose also that there is $L > 0$ such that for all $x, \bar{y}, \tilde{y}, h$,

$$|\Phi(x, \bar{y}; h) - \Phi(x, \tilde{y}; h)| \le L|\bar{y} - \tilde{y}|.$$

Let $Y_0 = y(x_0)$ and for $i = 1, \dots, N$,

$$Y_i = Y_{i-1} + h\Phi(x_{i-1}, Y_{i-1}; h).$$

Then for all $n = 0, 1, \dots, N$,

$$|y(x_n) - Y_n| \le \frac{T h^p}{L}(e^{nhL} - 1).$$

This theorem says that under mild conditions, the order of the global error is one less than that of the local truncation error.

Proof.

The result is proved by mathematical induction on n. For the base case, i.e. $n = 0$, the result is clearly true. Now as induction hypothesis suppose that the result holds when $n = i - 1 \geq 0$. We have

$$
\begin{aligned}
|y(x_i) - Y_i| &\leq |y(x_i) - y(x_{i-1}) - h\Phi(x_{i-1}, y(x_{i-1}); h)| + |y(x_{i-1}) + h\Phi(x_{i-1}, y(x_{i-1}); h) - Y_i| \\
&= |\tau_{i-1}| + |y(x_{i-1}) + h\Phi(x_{i-1}, y(x_{i-1}); h) - Y_{i-1} - h\Phi(x_{i-1}, Y_{i-1}; h)| \\
&\leq |\tau_{i-1}| + |y(x_{i-1}) - Y_{i-1}| + |h\Phi(x_{i-1}, y(x_{i-1}); h) - h\Phi(x_{i-1}, Y_{i-1}; h)| \\
&\leq Th^{p+1} + (1 + hL)|y(x_{i-1}) - Y_{i-1}| \\
&\leq Th^{p+1} + (1 + hL)\frac{Th^p}{L}(e^{(i-1)hL} - 1) \\
&= \frac{Th^p}{L}[(1 + hL)(e^{(i-1)hL} - 1) + hL],
\end{aligned}
$$

where the least inequality is obtained from the induction hypothesis.

Proof (continued).

$$(1+hL)(e^{(i-1)hL}-1)+hL=(1+hL)\sum_{k=1}^{\infty}\frac{(i-1)^k(hL)^k}{k!}+hL$$

$$=\sum_{k=1}^{\infty}\frac{(i-1)^k(hL)^k}{k!}+\sum_{k=1}^{\infty}\frac{(i-1)^k(hL)^{k+1}}{k!}+hL$$

$$=(i-1)hL+\sum_{k=2}^{\infty}\frac{(i-1)^k(hL)^k}{k!}+\sum_{k=2}^{\infty}\frac{(i-1)^{k-1}(hL)^k}{(k-1)!}+hL$$

$$=ihL+\sum_{k=2}^{\infty}\frac{[(i-1)^k+k(i-1)^{k-1}](hL)^k}{k!}$$

$$\leq ihL+\sum_{k=2}^{\infty}\frac{i^k(hL)^k}{k!}=e^{ihL}-1,$$

where the last inequality is based on a lemma that will be shown below. It then follows that

$$|y(x_i)-Y_i|\leq\frac{Th^p}{L}(e^{ihL}-1).$$

Lemma

If $i \geq 1$ and $k \geq 2$, then $(i-1)^k + k(i-1)^{k-1} < i^k$.

Proof.

If $i = 1$, then the result is clearly true. Now suppose $i > 1$, then

$$1 + \frac{k}{i-1} < (1 + \frac{1}{i-1})^k$$

$$\frac{i-1+k}{i-1} < (\frac{i}{i-1})^k$$

$$(i-1)^{k-1}(i-1+k) < i^k$$

$$(i-1)^k + k(i-1)^{k-1} < i^k.$$

□

Revisiting convergence of one-step methods

Let us now examine how fast is the convergence when there is error in computing Y_{i+1}. That is,

$$Y_{i+1} = Y_i + h\Phi(x_i, Y_i; h) + \varepsilon_i$$

where ε_i is the computing error.
Since we have used letter Y in error-less equation and derived theorems using it, we will retain the symbol Y and use a different letter Z for the error-contaminated version of Y. That is, we assume that

$$Z_0 = y_0 \tag{20}$$

$$Z_{i+1} = Z_i + h\Phi(x_i, Z_i; h) + \varepsilon_i. \tag{21}$$

We will now derive a theorem about the convergence of Z.

Revisiting convergence of one-step methods (continued)

Theorem

Suppose that there are $T > 0$ and $p > 0$ such that for $i = 0, 1, \dots, N-1$, $|\tau_i| \le T h^{p+1}$, and that there is $\Delta > 0$ such that for $i = 0, 1, \dots, N-1$, $|\varepsilon_i| \le h\Delta$. Suppose also that there is $L > 0$ such that for all $x, \bar{y}, \tilde{y}, h$,

$$|\Phi(x, \bar{y}; h) - \Phi(x, \tilde{y}; h)| \le L|\bar{y} - \tilde{y}|.$$

Let $Z_0 = y(x_0)$ and for $i = 1, \dots, N$,

$$Z_i = Z_{i-1} + h\Phi(x_{i-1}, Z_{i-1}; h) + \varepsilon_{i-1}.$$

Then for all $n = 0, 1, \dots, N$,

$$|y(x_n) - Z_n| \le \frac{T h^p + \Delta}{L}(e^{nhL} - 1).$$

We will explain this theorem after giving its proof.

Proof.

The result is proved by mathematical induction on n. For the base case, i.e. $n = 0$, the result is clearly true. Now as induction hypothesis suppose that the result holds when $n = i - 1 \geq 0$. We have

$$
\begin{aligned}
|y(x_i) - Z_i| &\leq |y(x_i) - y(x_{i-1}) - h\Phi(x_{i-1}, y(x_{i-1}); h)| + |y(x_{i-1}) + h\Phi(x_{i-1}, y(x_{i-1}); h) - Z_i| + |\varepsilon_{i-1}| \\
&= |\tau_{i-1}| + |\varepsilon_{i-1}| + |y(x_{i-1}) + h\Phi(x_{i-1}, y(x_{i-1}); h) - Y_{i-1} - h\Phi(x_{i-1}, Y_{i-1}; h)| \\
&\leq |\tau_{i-1}| + |\varepsilon_{i-1}| + |y(x_{i-1}) - Z_{i-1}| + |h\Phi(x_{i-1}, y(x_{i-1}); h) - h\Phi(x_{i-1}, Z_{i-1}; h)| \\
&\leq Th^{p+1} + h\Delta + (1 + hL)|y(x_{i-1}) - Z_{i-1}| \\
&\leq Th^{p+1} + h\Delta + (1 + hL)\frac{Th^p + \Delta}{L}(e^{(i-1)hL} - 1) \\
&= \frac{Th^p + \Delta}{L}[(1 + hL)(e^{(i-1)hL} - 1) + hL] \\
&= \frac{Th^p + \Delta}{L}(e^{ihL} - 1).
\end{aligned}
$$

□

Explaining the convergence theorem

With the mild assumption of Lipschitz condition of Φ, for a consistent method (its local truncation error decays faster than h), we have the following:

1. If both $h \to 0$ and $\Delta \to 0$, then the numerical solution converges to the exact solution.
2. If such a Δ exists but cannot be made to converge to 0, then there may be bounded discrepancy between the numerical solution and the exact solution, but the error is controlled.
3. Establishment of the convergence relies on the Lipschitz assumption of Φ.
4. If the computing error only has an absolute bound, i.e. one that does not decrease in proportion to h, then the global error may not be under control.

Order of a method

In practice, the computing error is negligible compared to the truncation error, so we now turn back to the error-less formula for a discussion of the order of a method.

If the global error of a method is $O(h^p)$, then the order of that method is p. It is one less than the order of its local truncation error.

From Euler to Runge-Kutta

Forward and backward Euler methods are only two simplest one-step methods. We now go on with deriving new methods, and finally we will arrive at a systematic approach of Runge-Kutta to derive the most advanced one-step methods.

The midpoint method

Again we start from

$$y(x_{i+1}) = y(x_i) + \int_{x_i}^{x_{i+1}} f(t, y(t))dt.$$

Applying the midpoint rule to the interval $[x_i, x_{i+1}]$, we obtain

$$y(x_{i+1}) \approx y(x_i) + hf(x_i + \frac{h}{2}, y(x_i + \frac{h}{2})).$$

Now the problem with using this expression to generate a scheme is that we do not know $y(x_i + \frac{h}{2})$. So what can we do? The only option is to approximate it there so an obvious approach is to take a step of length $\frac{h}{2}$ using forward Euler, i.e.,

$$y(x_i + \frac{h}{2}) \approx Y_i + \frac{h}{2} f(x_i, Y_i).$$

So the following equations describe the **midpoint method**:

$$
\begin{aligned}
K_1 &= f(x_i, Y_i) \\
K_2 &= f(x_i + \frac{h}{2}, Y_i + \frac{h}{2}K_1) \\
Y_{i+1} &= Y_i + hK_2.
\end{aligned}
$$

This method is clearly explicit because we are evaluating the slope at known points. When we look at the scheme we see that we actually have to perform two function evaluations whereas for the forward Euler we only had to evaluate $f(x_i, Y_i)$. Since we are doing more work we should expect better results and, in fact, this method is second order accurate which we will demonstrate later.

Local truncation error of the midpoint method

To avoid clutter, denote

$$\phi = f(x_i, y(x_i)), \quad \phi_1 = f_1(x_i, y(x_i)), \quad \phi_2 = f_2(x_i, y(x_i)).$$

We have

$$\begin{aligned} y(x_{i+1}) &= y(x_i) + hy'(x_i) + \frac{h^2}{2}y''(x_i) + O(h^3) \\ &= y(x_i) + h\phi + \frac{h^2}{2}(\phi_1 + \phi_2\phi) + O(h^3). \end{aligned}$$

The local truncation error at step i, denoted by τ_i, is

$$\begin{aligned} \tau_i &= y(x_{i+1}) - y(x_i) - hf(x_i + \frac{h}{2}, y(x_i) + \frac{h}{2}f(x_i, y(x_i))) \\ &= h\phi + \frac{h^2}{2}(\phi_1 + \phi_2\phi) + O(h^3) - h[\phi + \frac{h}{2}\phi_1 + \frac{h}{2}\phi_2\phi + O(h^2)] \\ &= h\phi + \frac{h^2}{2}(\phi_1 + \phi_2\phi) + O(h^3) - h\phi - \frac{h^2}{2}(\phi_1 + \phi_2\phi) + O(h^3) \\ &= O(h^3). \end{aligned}$$

The Trapezoid method

Again we start from

$$y(x_{i+1}) = y(x_i) + \int_{x_i}^{x_{i+1}} f(t, y(t))dt.$$

Applying the Trapezoid rule to the interval $[x_i, x_{i+1}]$, we obtain

$$y(x_{i+1}) \approx y(x_i) + \frac{h}{2}[f(x_i, y(x_i)) + f(x_{i+1}, y(x_{i+1}))].$$

So the following equations describe the **Trapezoid method**:

$$\begin{aligned} K &= f(x_i, Y_i) \\ Y_{i+1} &= Y_i + \frac{h}{2}[K + f(x_{i+1}, Y_{i+1})]. \end{aligned}$$

This method is also called the **modified Euler method**. It is clearly implicit because it requires evaluating the slope at the unknown point (x_{i+1}, Y_{i+1}).

Local truncation error of the modified Euler method

The local truncation error of the modified Euler method is also of order 3. Its proof is rather tricky so is omitted here.

So both the midpoint and Trapezoid methods have order 2.

Introducing Runge-Kutta methods

Runge and Kutta developed a systematic approach to derive high-order one-step methods for numerical solution of ODEs. In retrospect, even the forward and backward Euler methods, which was derived using quadrature rules before Runge-Kutta, can also be derived using the approach of Runge-Kutta. Therefore, it is somewhat controversial as to what methods are Runge-Kutta and what are not. In fact, all one-step methods can be classified as belonging to Runge-Kutta.

Runge-Kutta (RK) methods include both explicit and implicit methods. They are further characterized by the number of stages which is just the number of function evaluations performed at each time step.

The forward Euler and the midpoint methods are examples of explicit RK methods; the Euler method is a one-stage method whereas the midpoint method is a two-stage method because it uses information at the midpoint $x_i + \frac{h}{2}$ as well as at x_i. Both the backward Euler and the trapezoidal methods are examples of implicit RK methods; the backward Euler method is a one-stage method whereas the trapezoidal method is a two-stage method. The family of RK methods were developed primarily from 1895 to 1925 and involved work by Runge, Heun, Kutta and Nystrom.

Explicit Runge-Kutta methods of order 2

The standard approach for the derivation of families of RK methods is to use the method of undetermined coefficients because it gives families of methods with a prescribed local truncation error. Let us now consider the most general form of an explicit two-stage single step method below:

$$\begin{aligned} K_1 &= f(x_i, Y_i) \\ K_2 &= f(x_i + c_2 h, Y_i + h a_{21} K_1) \\ Y_{i+1} &= Y_i + h(b_1 K_1 + b_2 K_2). \end{aligned}$$

Assuming $Y_i = y(x_i)$, then as we have derived,

$$\begin{aligned} y(x_{i+1}) &= y(x_i) + h\phi + \frac{h^2}{2}(\phi_1 + \phi_2 \phi) + O(h^3) \\ &= Y_i + h\phi + \frac{h^2}{2}(\phi_1 + \phi_2 \phi) + O(h^3). \end{aligned}$$

The local truncation error at step i, denoted by τ_i, is

$$
\begin{aligned}
\tau_i &= y(x_{i+1}) - Y_i - h[b_1 f(x_i, Y_i) + b_2 f(x_i + c_2 h, Y_i + h a_{21} f(x_i, y(x_i)))] \\
&= h\phi + \frac{h^2}{2}(\phi_1 + \phi_2\phi) + O(h^3) - h[(b_1 + b_2)\phi + h b_2 c_2 \phi_1 + h b_2 a_{21} \phi_2 \phi + O(h^2)] \\
&= h\phi + \frac{h^2}{2}\phi_1 + \frac{h^2}{2}\phi_2\phi + O(h^3) - (b_1 + b_2)h\phi - b_2 c_2 h^2 \phi_1 - b_2 a_{21} h^2 \phi_2 \phi - O(h^3) \\
&= (1 - b_1 - b_2)h\phi + (\frac{1}{2} - b_2 c_2)h^2 \phi_1 + (\frac{1}{2} - b_2 a_{21})h^2 \phi_2 \phi + O(h^3).
\end{aligned}
$$

To allow $\tau_i = O(h^3)$, it must be satisfied that $b_1 + b_2 = 1$, $b_2 c_2 = \frac{1}{2}$, **and** $b_2 a_{21} = \frac{1}{2}$. The midpoint method is the case $b_1 = 0$, $b_2 = 1$, $c_2 = a_{21} = \frac{1}{2}$. There is also the case $b_1 = \frac{1}{2}$, $b_2 = \frac{1}{2}$, $c_2 = a_{21} = 1$, which is called the **improved Euler method**:

$$\begin{aligned} K_1 &= f(x_i, Y_i) \\ K_2 &= f(x_i + h, Y_i + hK_1) \\ Y_{i+1} &= Y_i + \frac{h}{2}(K_1 + K_2). \end{aligned}$$

And there is also the case $b_1 = \frac{1}{4}$, $b_2 = \frac{3}{4}$, $c_2 = a_{21} = \frac{2}{3}$, which is called the **Heun method**:

$$\begin{aligned} K_1 &= f(x_i, Y_i) \\ K_2 &= f(x_i + \frac{2}{3}h, Y_i + \frac{2}{3}hK_1) \\ Y_{i+1} &= Y_i + \frac{h}{4}(K_1 + 3K_2). \end{aligned}$$

Kutta's method

We now consider the most general form of an explicit three-stage single step method below:

$$\begin{aligned} K_1 &= f(x_i, Y_i) \\ K_2 &= f(x_i + c_2 h, Y_i + h a_{21} K_1) \\ K_3 &= f(x_i + c_3 h, Y_i + h(a_{31} K_1 + a_{32} K_2)) \\ Y_{i+1} &= Y_i + h(b_1 K_1 + b_2 K_2 + b_3 K_3). \end{aligned}$$

To find a 3rd-order method we need to derive a set of equations and solve it, which is very tricky for manual computation, and will be omitted here. One most-often used solution leads to the following **Kutta's method**:

$$\begin{aligned} K_1 &= f(x_i, Y_i) \\ K_2 &= f(x_i + \frac{h}{2}, Y_i + \frac{h}{2} K_1) \\ K_3 &= f(x_i + h, Y_i + h(-K_1 + 2K_2)) \\ Y_{i+1} &= Y_i + \frac{h}{6}(K_1 + 4K_2 + K_3). \end{aligned}$$

The 4th-order classic Runge-Kutta method

The **classic Runge-Kutta method** is a 4th-order explicit method. Its detailed derivation is very complicated and is also omitted. It is described as follows:

$$
\begin{aligned}
K_1 &= f(x_i, Y_i) \\
K_2 &= f(x_i + \frac{h}{2}, Y_i + \frac{h}{2}K_1) \\
K_3 &= f(x_i + \frac{h}{2}, Y_i + \frac{h}{2}K_2) \\
K_4 &= f(x_i + h, Y_i + hK_3)) \\
Y_{i+1} &= Y_i + \frac{h}{6}(K_1 + 2K_2 + 2K_3 + K_4).
\end{aligned}
$$

The 2-stage 4th-order Gauss method

The 2-stage **Gauss method** is an implicit method and has order 4. Its detailed derivation is very complicated and is also omitted. It is described as follows:

$$
\begin{aligned}
K_1 &= f(x_i + (\frac{1}{2} - \frac{\sqrt{3}}{6})h, Y_i + h(\frac{1}{4}K_1 + (\frac{1}{4} - \frac{\sqrt{3}}{6})K_2)) \\
K_2 &= f(x_i + (\frac{1}{2} + \frac{\sqrt{3}}{6})h, Y_i + h((\frac{1}{4} + \frac{\sqrt{3}}{6})K_1 + \frac{1}{4}K_2)) \\
Y_{i+1} &= Y_i + \frac{h}{2}(K_1 + K_2).
\end{aligned}
$$

The 2-stage 3th-order Radau method

The 2-stage **Radau method** is an implicit method and has order 3. Its detailed derivation is very complicated and is also omitted. It is described as follows:

$$
\begin{aligned}
K_1 &= f(x_i + \frac{1}{3}h, Y_i + h(\frac{5}{12}K_1 - \frac{1}{12}K_2)) \\
K_2 &= f(x_i + h, Y_i + h(\frac{3}{4}K_1 + \frac{1}{4}K_2)) \\
Y_{i+1} &= Y_i + h(\frac{3}{4}K_1 + \frac{1}{4}K_2).
\end{aligned}
$$

The 3-stage 5th-order Radau method

The 3-stage **Radau method** is an implicit method and has order 5. Its detailed derivation is very complicated and is also omitted. It is described as follows:

$$
\begin{aligned}
K_1 &= f(x_i, Y_i + h(\frac{1}{9}K_1 + \frac{-1-\sqrt{6}}{18}K_2 + \frac{-1+\sqrt{6}}{18}K_3)) \\
K_2 &= f(x_i + \frac{6-\sqrt{6}}{10}h, Y_i + h(\frac{1}{9}K_1 + \frac{88+7\sqrt{6}}{360}K_2 + \frac{88-43\sqrt{6}}{360}K_3)) \\
K_3 &= f(x_i + \frac{6+\sqrt{6}}{10}h, Y_i + h(\frac{1}{9}K_1 + \frac{88+43\sqrt{6}}{360}K_2 + \frac{88-7\sqrt{6}}{360}K_3)) \\
Y_{i+1} &= Y_i + h(\frac{1}{9}K_1 + \frac{16+\sqrt{6}}{36}K_2 + \frac{16-\sqrt{6}}{36}K_3).
\end{aligned}
$$

Stability analysis of one-step methods

We have known that all one-step methods have zero-stability, i.e., any one-step method can be made stable by choosing a sufficiently small step size h. But how small should h be to make it stable? It turns out that for general nonlinear f, determining a suitable step size so that the method is stable is very complicated. Therefore, it is a common practice to use linear ordinary differential equations (ODEs) like

$$y' = \lambda y, \text{ where } \lambda \text{ is a complex number with } \mathrm{Re}(\lambda) < 0$$

to test the stability of a numerical solution method.

Why use linear functions?

Because it is simple to analyze, while still allowing us to have a full understanding of the stability behavior of numeric solution methods.

Why allow λ to be complex?

By allowing λ to be complex, we can explore a wider range of behaviors for the solutions. The imaginary part of λ, introduces oscillatory behavior in the solutions. This is particularly relevant for systems that can exhibit cyclical or oscillatory phenomena.

Why consider only λ on the left-half plane?

If $\mathrm{Re}(\lambda) > 0$, then the system itself is unstable, and then it does not matter whether the numeric method is stable or not.

Stability region of the forward Euler method

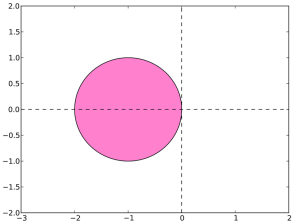
The forward Euler method is described by

$$Y_{i+1} = Y_i + hf(x_i, Y_i).$$

Consider the test equation $y'(x) = \lambda y(x)$ with $\mathrm{Re}(\lambda) < 0$, and apply the method to the test equation. We have

$$Y_{i+1} = Y_i + h\lambda Y_i,$$
$$Y_{i+1} = (1 + h\lambda)Y_i.$$

Let $z = h\lambda$. The stability region is the region of the complex plane where $|1 + z| < 1$. This region is the disk centered at $(-1, 0)$ with radius 1.



Stability region of the backward Euler method

The backward Euler method is described by

$$Y_{i+1} = Y_i + hf(x_{i+1}, Y_{i+1}).$$

Consider the test equation $y'(x) = \lambda y(x)$ with $\mathrm{Re}(\lambda) < 0$, and apply the method to the test equation. We have

$$Y_{i+1} = Y_i + h\lambda Y_{i+1},$$

$$Y_{i+1} = (1 - h\lambda)^{-1} Y_i.$$

Let $z = h\lambda$. The stability region is the region of the complex plane where $|(1 - z)^{-1}| < 1$.

This is equivalent to $|1 - z| > 1$. It is apparent that the stability region includes the entire left half plane.

Multistep Methods

Introducing linear multistep methods

The methods discussed to this point in the chapter are called one-step methods because the approximation for the mesh point x_{i+1} involves information from only one of the previous mesh points, x_i. Although these methods might use function evaluation information at points between x_i and x_{i+1}, they do not retain that information for direct use in future approximations. All the information used by these methods is obtained within the subinterval over which the solution is being approximated.

The approximate solution is available at each of the mesh points $x_0, x_1, \ldots, x_i$ before the approximation at x_{i+1} is obtained, and because the error $|y(x_i) - Y_i|$ tends to increase with i, so it seems reasonable to develop methods that use these more accurate previous data when approximating the solution at x_{i+1}. Methods using the approximation at more than one previous mesh point to determine the approximation at the next point are called **linear multistep methods**.

Definition of linear multistep methods

From now on, let us denote $F_i = f(x_i, Y_i)$ for $i = 0, 1, \ldots, N$.

Definition

A general linear m-step method has the form

$$Y_{i+1} = \sum_{j=1}^{m} \alpha_j Y_{i+1-j} + h \sum_{j=0}^{m} \beta_j F_{i+1-j}. \tag{22}$$

If $\beta_0 = 0$ then the method is explicit; otherwise it is implicit. We don't include a term $\alpha_0 Y_{i+1}$ on the right-hand side for implicit methods because if we did we could just combine it with the Y_{i+1} term on the left-hand side of the formula and then divide by that coefficient to get the form above.

The integer m is the number of steps of this method, but not necessarily the order of the method. The order is defined in terms of its truncation error.

Convergence of a LMM

Definition

A numeric method (22) is said to converge to IVODE (1), if with assumption (3),

$$\lim_{h\to 0, Y_0\to y(x_0),\ldots,Y_{m-1}\to y(x_{m-1})} \max_{m\le i\le N} |y(x_i) - Y_i| = 0. \tag{23}$$

We do not suppose that the scheme has the exact initial condition of the ODE. Indeed, on one hand, there could be truncation errors in the initial conditions. On the other hand, the initial conditions may not be known exactly.

Generally, just as the solution of a differential system is continuous with respect to the initial data, the solution approximated by a multistep scheme must be continuous with respect to a perturbation of the initial conditions.

Convergence, as we will see, results from two properties. The first, stability, is a property of the scheme. This ensures that the scheme does not amplify too much the errors created at each step. The other, consistency, describes a relation between the scheme and the differential system. It implies that the scheme does not differ much from the solution locally.

Zero-stability of a LMM

Definition

A numeric method (22) is said to be zero-stable if with assumption (3), there exist h^* and a constant M such that, for all $Y_0, \ldots, Y_{m-1}$, for all $Z_0, \ldots, Z_{m-1}$, for all $h \leq h^*$, and for every sequence of vectors ϵ_i, the sequences Y_i and Z_i defined by the relations for $i \geq m-1$,

$$\begin{aligned} Y_{i+1} &= \sum_{j=1}^{m} \alpha_j Y_{i+1-j} + h \sum_{j=0}^{m} \beta_j f(x_{i+1-j}, Y_{i+1-j}) \\ Z_{i+1} &= \sum_{j=1}^{m} \alpha_j Z_{i+1-j} + h \sum_{j=0}^{m} \beta_j f(x_{i+1-j}, Z_{i+1-j}) + \epsilon_i \end{aligned} \tag{24}$$

satisfy the inequality

$$|Y_i - Z_i| \leq M\Big(\sum_{k=0}^{m-1} |Y_k - Z_k| + \sum_{k=m-1}^{i-1} |\epsilon_k|\Big), \quad \forall m \leq i \leq N(h). \tag{25}$$

Local truncation error of an LMM

For an IVODE (1–3) and a linear multistep method (15), the local truncation error at the i-th step is defined as

$$\tau_i = y(x_{i+1}) - Y_{i+1} \tag{26}$$

assuming that $Y_i = y(x_i), Y_{i-1} = y(x_{i-1}), \cdots, Y_{i+1-m} = y(x_{i+1-m})$.

Again, this is a mathematical concept and does not exist in practice.

Consistency of a LMM

Definition (consistency)

Numeric method (22) is said to be consistent with (1), if with assumption (3),

$$\lim_{h\to 0}\sum_{i=m-1}^{N(h)-1}|\tau_i| = 0. \tag{27}$$

Convergence = zero stability + consistency

Theorem

A numeric method (22) converges to IVODE (1), if it is zero-stable and consistent with (1).

A sufficient condition for zero stability

Theorem

For method (22), suppose

- *There exists h^* and a constant L such that*

$$|f(x,y) - f(x,z)| \le L|y-z|, \quad \forall x \in [a,b], \forall y, \forall z, \forall h \in (0,h^*]. \tag{28}$$

- *The polynomial p, defined by*

$$p(x) = x^m - \sum_{k=1}^{m} \alpha_k x^{m-k} \tag{29}$$

has all of its roots in the unit disk, while its roots of modulus 1 are simple.

Then the method (22) is zero-stable.

How to start a linear multistep method?

A drawback of an m-step ($m > 1$) method is that we need m starting values $Y_0, Y_1, \cdots, Y_{m-1}$ and we only have Y_0 from the initial condition. Typically one uses a one-step method to start the scheme. How do we decide what method to use? A "safe" approach is to use a method which has the same accuracy as the multistep method (e.g., a Runge-Kutta method).

LMM example – modified midpoint method

We start from

$$y(x_{i+1}) = y(x_{i-1}) + \int_{x_{i-1}}^{x_{i+1}} f(t, y(t))dt.$$

Now if we use the midpoint rule to approximate the integral on the right we have the two-step scheme:

$$Y_{i+1} = Y_{i-1} + 2hf(x_i, Y_i).$$

This is sometimes called the **modified midpoint method**.

What methods are simultaneously Runge-Kutta and linear multistep?

An m-step method that is simultaneously Runge-Kutta must have $m = 1$, i.e. only $\alpha_1, \beta_0, \beta_1$ can be nonzero. That is,

$$Y_{i+1} = \alpha_1 Y_i + h[\beta_0 f(x_{i+1}, Y_{i+1}) + \beta_1 f(x_i, Y_i)].$$

We now consider its local truncation error. If $\beta_0 \neq 0$, then the method is implicit, and further derivation becomes very tricky, and will be omitted here. We now assume $\beta_0 = 0$, then, assuming $Y_i = y(x_i)$, we have

$$\begin{aligned}
\tau_i &= y(x_{i+1}) - Y_{i+1} \\
&= y(x_{i+1}) - \alpha_1 Y_i - h\beta_1 f(x_i, Y_i) \\
&= y(x_{i+1}) - \alpha_1 y(x_i) - h\beta_1 f(x_i, y(x_i)) \\
&= y(x_i + h) - \alpha_1 y(x_i) - h\beta_1 y'(x_i) \\
&= y(x_i) + hy'(x_i) + O(h^2) - \alpha_1 y(x_i) - h\beta_1 y'(x_i) \\
&= (1 - \alpha_1) y(x_i) + h(1 - \beta_1) y'(x_i) + O(h^2).
\end{aligned}$$

For this explict method to be at least order 1, it must be satisfied that $\alpha_1 = 1$, $\beta_1 = 1$.

What methods are simultaneously Runge-Kutta and multistep? (continued)

In fact, if the implicit case is also included, then for the method to be at least order 1, it must be satisfied that $\alpha_1 = 1$, $\beta_0 + \beta_1 = 1$. That is:

$$Y_{i+1} = Y_i + h[\beta_0 F_{i+1} + (1 - \beta_0) F_i].$$

They include the following well-known methods:

1. Forward Euler ($\beta_0 = 0$)
2. Backward Euler ($\beta_0 = 1$)
3. Modified Euler (Trapezoid, $\beta_0 = \frac{1}{2}$)

In addition, the Trapezoid method is the only such method that is order 2. Other choices of β_0 are useless, because they are definitely inferior to one of the above three methods.

Derivation of general 2-step methods

We want to show how linear multistep methods are derived and how to decide which ones will work best. The main issues that arise with multistep methods can be introduced in the relatively simple case of two-step methods, so we begin there. A general two-step method has the form

$$Y_{i+1} = \alpha_1 Y_i + \alpha_2 Y_{i-1} + h(\beta_0 F_{i+1} + \beta_1 F_i + \beta_2 F_{i-1}).$$

To develop a linear multistep method, we need to refer to Taylor's Theorem, since we try to match as many terms of the solution's Taylor expansion as possible with the terms of the method. What remains will be the local truncation error.

We assume that all previous Y's are correct – that is, $Y_i = y(x_i)$ and $Y_{i-1} = y(x_{i-1})$. The differential equation says that $y'(x_i) = F_i$, so all terms can be expanded in a Taylor expansion as follows:

$$
\begin{aligned}
Y_{i+1} &= \alpha_1 y(x_i) + \alpha_2 y(x_{i-1}) + h[\beta_0 y'(x_{i+1}) + \beta_1 y'(x_i) + \beta_2 y'(x_{i-1})] \\
&= \alpha_1 y(x_i) + \alpha_2 y(x_i - h) + h[\beta_0 y'(x_i + h) + \beta_1 y'(x_i) + \beta_2 y'(x_i - h)] \\
&= \alpha_1 y(x_i) \\
&\quad +\alpha_2[y(x_i) \quad -hy'(x_i) \quad +\frac{h^2}{2}y''(x_i) \quad -\frac{h^3}{6}y'''(x_i) \quad +\frac{h^4}{24}y^{(4)}(x_i) \quad -\cdots] \\
&\quad +\beta_0[\quad hy'(x_i) \quad +h^2 y''(x_i) \quad +\frac{h^3}{2}y'''(x_i) \quad +\frac{h^4}{6}y^{(4)}(x_i) \quad +\cdots] \\
&\quad +\beta_1[\quad hy'(x_i)] \\
&\quad +\beta_2[\quad hy'(x_i) \quad -h^2 y''(x_i) \quad +\frac{h^3}{2}y'''(x_i) \quad -\frac{h^4}{6}y^{(4)}(x_i) \quad +\cdots].
\end{aligned}
$$

Adding up yields

$$
Y_{i+1} = (\alpha_1 + \alpha_2)y(x_i) + (-\alpha_2 + \beta_0 + \beta_1 + \beta_2)hy'(x_i) + (\alpha_2 + 2\beta_0 - 2\beta_2)\frac{h^2}{2}y''(x_i)
$$

$$
+(-\alpha_2 + 3\beta_0 + 3\beta_2)\frac{h^3}{6}y'''(x_i) + (\alpha_2 + 4\beta_0 - 4\beta_2)\frac{h^4}{24}y^{(4)}(x_i) + \cdots .
$$

By choosing the α_j and β_j appropriately, the local truncation error $y(x_{i+1}) - Y_{i+1}$, where

$$
y(x_{i+1}) = y(x_i) + hy'(x_i) + \frac{h^2}{2}y''(x_i) + \frac{h^3}{6}y'''(x_i) + \frac{h^4}{24}y^{(4)}(x_i) + \cdots
$$

can be made as small as possible, assuming that the derivatives involved actually exist. Next, we will investigate the possibilities.

To look for explicit methods, set $\beta_0 = 0$. So there are four undetermined coefficients, and it seems possible to obtain a third-order method. But as we will see later, this third-order method is unstable and is not usable at all. So we have to content ourselves with a second-order method. It can be developed by matching terms in the two preceding equations up to and including the h^2 term, making the local truncation error of size $O(h^3)$. Comparing terms yields the system

$$\begin{aligned} \alpha_1 + \alpha_2 &= 1 \\ -\alpha_2 + \beta_1 + \beta_2 &= 1 \\ \alpha_2 - 2\beta_2 &= 1 \end{aligned}$$

There are three equations in four unknowns $\alpha_1, \alpha_2, \beta_1, \beta_2$, so it will be possible to find infinitely many different explicit order-two methods. Note that the equations can be written in terms of α_1 as follows:

$$\begin{aligned} \alpha_2 &= 1 - \alpha_1 \\ \beta_1 &= 2 - \frac{1}{2}\alpha_1 \\ \beta_2 &= -\frac{1}{2}\alpha_1. \end{aligned}$$

The local truncation error will be

$$y(x_{i+1}) - Y_{i+1} = \frac{4+\alpha_1}{12} h^3 y'''(x_i) + O(h^4).$$

We are free to set α_1 arbitrarily – any choice leads to a second-order method, as we have just shown. Setting $\alpha_1 = 1$ yields the second-order Adams-Bashforth Method. Note that $\alpha_2 = 0$ by the first equation, and $\beta_2 = -1/2$ and $\beta_1 = 3/2$. The local truncation error is $\frac{5}{12} h^3 y'''(x_i) + O(h^4)$.

Alternatively, we could set $\alpha_1 = 1/2$ to get another two-step second-order method with $\alpha_2 = 1/2$, $\beta_1 = 7/4$, and $\beta_2 = -1/4$:

$$Y_{i+1} = \frac{1}{2} Y_i + \frac{1}{2} Y_{i-1} + h[\frac{7}{4} F_i - \frac{1}{4} F_{i-1}].$$

This method has local truncation error $\frac{3}{8} h^3 y'''(x_i) + O(h^4)$.

Definition of Adams methods

Definition

An m-step method is called an **Adams method** if $\alpha_j = 0$ for $j \neq 1$.

That means, in an Adams method, in the α coefficients, only α_1 can be nonzero, which must be 1:

$$Y_{i+1} = Y_i + h \sum_{j=0}^{m} \beta_j F_{i+1-j}.$$

Adams methods are further subdivided into Adams-Bashforth methods and Adams-Moulton methods.

Definition

An Adams method is called an **Adams-Bashforth** method if $\beta_0 = 0$, and an **Adams-Moulton** method otherwise.

Derivation of Adams-Bashforth formulas

Let us start from the general Adams-Bashforth formula:

$$Y_{i+1} = Y_i + h \sum_{j=1}^{m} \beta_j F_{i+1-j}.$$

We assume that all previous Y's are correct, so all terms can be expanded in a Taylor expansion as follows:

$$\begin{aligned}
Y_{i+1} &= y(x_i) + h\beta_1 y'(x_i) + h \sum_{j=1}^{m-1} \beta_{j+1} y'(x_i - jh) \\
&= y(x_i) + h\beta_1 y'(x_i) + h \sum_{j=1}^{m-1} \beta_{j+1} \sum_{k=0}^{\infty} y^{(k+1)}(x_i) \frac{(-j)^k h^k}{k!} \\
&= y(x_i) + h\beta_1 y'(x_i) + \sum_{k=0}^{\infty} \left(\sum_{j=1}^{m-1} \beta_{j+1} (-j)^k \, y^{(k+1)}(x_i) \frac{h^{k+1}}{k!} \right) \\
&= y(x_i) + \sum_{j=1}^{m} \beta_j y'(x_i) h + \sum_{k=2}^{\infty} \left(\sum_{j=1}^{m-1} \beta_{j+1} (-j)^{k-1} \, y^{(k)}(x_i) \frac{h^k}{(k-1)!} \right).
\end{aligned}$$

Note that

$$y(x_{i+1}) = y(x_i) + \sum_{k=1}^{\infty} y^{(k)}(x_i)\frac{h^k}{k!}.$$

Since we have m unknowns, we can match the h^k terms for $k = 1, \dots, m$, thus we obtain the following:

$$\sum_{j=1}^{m} \beta_j = 1,$$

$$\sum_{j=2}^{m} \beta_j (1-j)^{k-1} = \frac{1}{k}, \textbf{ for } k = 2, \dots, m.$$

The local truncation error is

$$\tau_i = [\frac{1}{(m+1)!} - \frac{\sum_{j=2}^{m}(1-j)^m \beta_j}{m!}] y^{(m+1)}(\xi_i) h^{m+1} \text{ where } \xi_i \in (x_{i+1} - mh, x_{i+1}).$$

Solving for the equations we obtain the following table.

Coefficients and local truncation error for Adams–Bashforth methods

m	β_1	β_2	β_3	β_4	β_5	β_6	β_7	β_8	τ
1	1								$\frac{1}{2}y''(\xi)h^2$
2	$\frac{3}{2}$	$-\frac{1}{2}$							$\frac{5}{12}y'''(\xi)h^3$
3	$\frac{23}{12}$	$-\frac{4}{3}$	$\frac{5}{12}$						$\frac{3}{8}y^{(4)}(\xi)h^4$
4	$\frac{55}{24}$	$-\frac{59}{24}$	$\frac{37}{24}$	$-\frac{3}{8}$					$\frac{251}{720}y^{(5)}(\xi)h^5$
5	$\frac{1901}{720}$	$-\frac{1387}{360}$	$\frac{109}{30}$	$-\frac{637}{360}$	$\frac{251}{720}$				$\frac{95}{288}y^{(6)}(\xi)h^6$
6	$\frac{4277}{1440}$	$-\frac{2641}{480}$	$\frac{4991}{720}$	$-\frac{3649}{720}$	$\frac{959}{480}$	$-\frac{95}{288}$			$\frac{19087}{60480}y^{(7)}(\xi)h^7$
7	$\frac{198721}{60480}$	$-\frac{18637}{2520}$	$\frac{235183}{20160}$	$-\frac{10754}{945}$	$\frac{135713}{20160}$	$-\frac{5603}{2520}$	$\frac{19087}{60480}$		$\frac{5257}{17280}y^{(8)}(\xi)h^8$
8	$\frac{16083}{4480}$	$-\frac{1152169}{120960}$	$\frac{242653}{13440}$	$-\frac{296053}{13440}$	$\frac{2102243}{120960}$	$-\frac{115747}{13440}$	$\frac{32863}{13440}$	$-\frac{5257}{17280}$	$\frac{1070017}{3628800}y^{(9)}(\xi)h^9$

Derivation of Adams-Moulton formulas

Let us start from the general Adams-Moulton formula:

$$Y_{i+1} = Y_i + h \sum_{j=0}^{m} \beta_j F_{i+1-j}.$$

We assume that all previous Y's are correct, so all terms can be expanded in a Taylor expansion as follows:

$$Y_{i+1} = y(x_{i+1} - h) + h \sum_{j=0}^{m} \beta_j y'(x_{i+1} - jh)$$

$$= y(x_{i+1}) + \sum_{k=1}^{\infty} y^{(k)}(x_{i+1}) \frac{(-h)^k}{k!} + h\beta_0 y'(x_{i+1}) + h \sum_{j=1}^{m} \beta_j \sum_{k=0}^{\infty} y^{(k+1)}(x_{i+1}) \frac{(-j)^k h^k}{k!}$$

$$= y(x_{i+1}) + h\beta_0 y'(x_{i+1}) + \sum_{k=1}^{\infty} y^{(k)}(x_{i+1}) \frac{(-h)^k}{k!} + \sum_{j=1}^{m} \beta_j \sum_{k=1}^{\infty} y^{(k)}(x_{i+1}) \frac{(-j)^{k-1} h^k}{(k-1)!}$$

$$= y(x_{i+1}) + h(\sum_{j=0}^{m} \beta_j - 1)y'(x_{i+1}) + \sum_{k=2}^{\infty} \left((-1)^k [1 - k \sum_{j=1}^{m} j^{k-1} \beta_j] \, y^{(k)}(x_{i+1}) \frac{h^k}{k!} \right)$$

Since we have $m+1$ unknowns, we can match the h^k terms for $k = 1, \ldots, m+1$, thus we obtain the following:

$$\sum_{j=0}^{m} \beta_j = 1,$$

$$\sum_{j=1}^{m} \beta_j j^{k-1} = \frac{1}{k}, \textbf{ for } k = 2, \ldots, m+1.$$

The local truncation error is

$$\tau_i = (-1)^m [\frac{\sum_{j=1}^{m} j^{m+1} \beta_j}{(m+1)!} - \frac{1}{(m+2)!}] y^{(m+2)}(\xi_i) h^{m+2} \text{ where } \xi_i \in (x_{i+1} - mh, x_{i+1}).$$

Solving for the equations we obtain the following table.

Coefficients and local truncation error for Adams–Moulton methods

m	β_0	β_1	β_2	β_3	β_4	β_5	β_6	β_7	τ
0	1								$-\frac{1}{2}y''(\xi)h^2$
1	$\frac{1}{2}$	$\frac{1}{2}$							$-\frac{1}{12}y'''(\xi)h^3$
2	$\frac{5}{12}$	$\frac{2}{3}$	$-\frac{1}{12}$						$-\frac{1}{24}y^{(4)}(\xi)h^4$
3	$\frac{3}{8}$	$\frac{19}{24}$	$-\frac{5}{24}$	$-\frac{1}{24}$					$-\frac{19}{720}y^{(5)}(\xi)h^5$
4	$\frac{251}{720}$	$\frac{323}{360}$	$-\frac{11}{30}$	$\frac{53}{360}$	$-\frac{19}{720}$				$-\frac{3}{160}y^{(6)}(\xi)h^6$
5	$\frac{95}{288}$	$\frac{1427}{1440}$	$-\frac{133}{240}$	$\frac{241}{720}$	$-\frac{173}{1440}$	$\frac{3}{160}$			$-\frac{863}{60480}y^{(7)}(\xi)h^7$
6	$\frac{19087}{60480}$	$\frac{2713}{2520}$	$-\frac{15487}{20160}$	$\frac{586}{945}$	$-\frac{6737}{20160}$	$\frac{263}{2520}$	$-\frac{863}{60480}$		$-\frac{275}{24192}y^{(8)}(\xi)h^8$
7	$\frac{5257}{17280}$	$\frac{139849}{120960}$	$-\frac{4511}{4480}$	$\frac{123133}{120960}$	$-\frac{88547}{120960}$	$\frac{1537}{4480}$	$-\frac{11351}{120960}$	$\frac{275}{24192}$	$-\frac{33953}{3628800}y^{(9)}(\xi)h^9$

It is remarkable that an m-step Adams-Bashforth method is of order m, whereas an m-step Adams-Moulton method is of order $m + 1$.

Definition of BDF methods

Definition

An m-step method is called a **BDF** method if $\beta_j = 0$ for $j \neq 0$.

That means, in a BDF method, in the β coefficients, only β_0 can be nonzero:

$$Y_{i+1} = \sum_{j=1}^{m} \alpha_j Y_{i+1-j} + h\beta_0 F_{i+1}.$$

Therefore all BDF methods are implicit. Can we define "explicit BDF" methods by stipulating that only β_1 can be nonzero in the β coefficients? No. Because we can do nothing with such a configuration. See the previous derivation of two-step methods.

Derivation of BDF formulas

Let us start from the general BDF formula:

$$Y_{i+1} = \sum_{j=1}^{m} \alpha_j Y_{i+1-j} + h\beta_0 F_{i+1}.$$

We assume that all previous Y's are correct, so all terms can be expanded in a Taylor expansion as follows:

$$Y_{i+1} = \sum_{j=1}^{m} \alpha_j y(x_{i+1} - jh) + h\beta_0 y'(x_{i+1})$$

$$= \sum_{j=1}^{m} \alpha_j \sum_{k=0}^{\infty} y^{(k)}(x_{i+1}) \frac{(-j)^k h^k}{k!} + h\beta_0 y'(x_{i+1})$$

$$= \sum_{j=1}^{m} \alpha_j y(x_{i+1}) + [\beta_0 - \sum_{j=1}^{m} j\alpha_j] y'(x_{i+1}) h + \sum_{k=2}^{\infty} \sum_{j=1}^{m} (-j)^k \alpha_j y^{(k)}(x_{i+1}) \frac{h^k}{k!}$$

Since we have $m+1$ unknowns, we can match the h^k terms for $k = 0, \dots, m$, thus we obtain the following:

$$\sum_{j=1}^{m} \alpha_j = 1,$$

$$\sum_{j=1}^{m} j^k \alpha_j = 0, \textbf{ for } k = 2, \dots, m,$$

$$\sum_{j=1}^{m} j \alpha_j = \beta_0.$$

The local truncation error is

$$\tau_i = (-1)^m \frac{\sum_{j=1}^{m} j^{m+1} \alpha_j}{(m+1)!} y^{(m+1)}(\xi_i) h^{m+1} \text{ where } \xi_i \in (x_{i+1} - mh, x_{i+1}).$$

Solving for the equations we obtain the following table.

Coefficients and local truncation error for BDF methods

m	β_0	α_1	α_2	α_3	α_4	α_5	α_6	τ
1	1	1						$-\frac{1}{2}y''(\xi)h^2$
2	$\frac{2}{3}$	$\frac{4}{3}$	$-\frac{1}{3}$					$-\frac{2}{9}y'''(\xi)h^3$
3	$\frac{6}{11}$	$\frac{18}{11}$	$-\frac{9}{11}$	$\frac{2}{11}$				$-\frac{3}{22}y^{(4)}(\xi)h^4$
4	$\frac{12}{25}$	$\frac{48}{25}$	$-\frac{36}{25}$	$\frac{16}{25}$	$-\frac{3}{25}$			$-\frac{12}{125}y^{(5)}(\xi)h^5$
5	$\frac{60}{137}$	$\frac{300}{137}$	$-\frac{300}{137}$	$\frac{200}{137}$	$-\frac{75}{137}$	$\frac{12}{137}$		$-\frac{10}{137}y^{(6)}(\xi)h^6$
6	$\frac{60}{147}$	$\frac{360}{147}$	$-\frac{450}{147}$	$\frac{400}{147}$	$-\frac{225}{147}$	$\frac{72}{147}$	$-\frac{10}{147}$	$-\frac{20}{343}y^{(7)}(\xi)h^7$